



Program/s: B Tech

Year: II Semester: IV

Stream/s: Artificial Intelligence

Subject: Machine Learning

Time: 3 hrs (10:00am to 1:00pm)Date: 24/04/2023

No. of Pages: 5

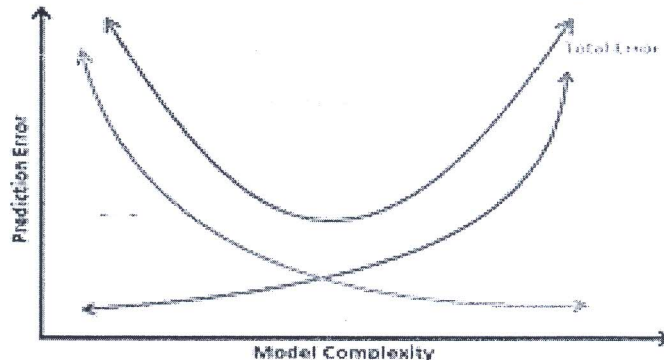
Marks: 100

Final Examination / Re Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) **In all 5 questions to be attempted.**
- 4) All questions carry equal marks.
- 5) **Answer to each new question to be started on a fresh page.**
- 6) **Figures in brackets on the right hand side indicate full marks.**
- 7) **Assume Suitable data if necessary.**

Q1		Answer briefly: (Each question carries 4 marks)	[20]
CO-1; SO-2; BL-2	a.	Why do we use descriptive and inferential statistics in machine learning data preparation, explain using example when needed? What is stratified sampling?	
CO-3; SO-2; BL-2	b.	Why do decision trees overfit, how can we overcome the overfitting of decision trees? (use schematics where possible)	
CO-3; SO-2; BL-4	c.	For SVM, derive the optimization equation used with the help of Lagrange multiplier. What will change in your output equation if we had non-linear decision boundaries?	
CO-1; SO-2; BL-2	d.	For the shown figure, please indicate the variance, Bias curves. Indicate where the model will overfit and underfit. Mention solution to overcome the overfitting and underfitting issues.	



CO-3; SO-2; BL-3	e.	Calculate the Entropy of hair color for the below, what would be the maximum Entropy in this case? <table><tr><th>Hair color</th><th>Count</th></tr><tr><td>Black</td><td>75</td></tr><tr><td>Brown</td><td>15</td></tr><tr><td>Blond</td><td>5</td></tr><tr><td>Red</td><td>0</td></tr><tr><td>Other</td><td>5</td></tr></table>	Hair color	Count	Black	75	Brown	15	Blond	5	Red	0	Other	5																																																																															
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Q2 CO-3; SO-3; BL-3		a. What is Gain Ratio? Why is Gain Ratio important? b. What is Bootstrap aggregation? What is Adaptive boosting? c. For the below Data set build a XGBoost classifier, use learning rate=1 and lambda=0, gamma=2, maximum depth=2. Build the M0 and M1 learners only. <table><tr><th>Dosage</th><th>Gender</th><th>Effect</th></tr><tr><td>2</td><td>M</td><td>0</td></tr><tr><td>8</td><td>F</td><td>1</td></tr><tr><td>12</td><td>M</td><td>1</td></tr><tr><td>18</td><td>F</td><td>0</td></tr></table>	Dosage	Gender	Effect	2	M	0	8	F	1	12	M	1	18	F	0	[20]																																																																											
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Q3 CO-3; SO-3; BL-3		a. What is K-fold cross validation? Why is it better than repeated hold-out method? b. For the below data set build a one split regression tree. Draw the one-split tree. c. For the second split use only the left most branch from previous split, draw the 2-split tree. d. What will be the output prediction from the regression tree if you had an input of (Outlook=Sunny, Temp.=Hot, Humidity=High, Wind=Weak) <table><tr><th>Day`</th><th>Outlook</th><th>Temp</th><th>Humidity</th><th>Wind</th><th>Golf Players</th></tr><tr><td>1</td><td>Sunny</td><td>Hot</td><td>High</td><td>Weak</td><td>25</td></tr><tr><td>2</td><td>Sunny</td><td>Hot</td><td>High</td><td>Strong</td><td>30</td></tr><tr><td>3</td><td>Overcast</td><td>Hot</td><td>High</td><td>Weak</td><td>46</td></tr><tr><td>4</td><td>Rain</td><td>Mild</td><td>High</td><td>Weak</td><td>45</td></tr><tr><td>5</td><td>Rain</td><td>Cool</td><td>Normal</td><td>Weak</td><td>52</td></tr><tr><td>6</td><td>Rain</td><td>Cool</td><td>Normal</td><td>Strong</td><td>23</td></tr><tr><td>7</td><td>Overcast</td><td>Cool</td><td>Normal</td><td>Strong</td><td>43</td></tr><tr><td>8</td><td>Sunny</td><td>Mild</td><td>High</td><td>Weak</td><td>35</td></tr><tr><td>9</td><td>Sunny</td><td>Cool</td><td>Normal</td><td>Weak</td><td>38</td></tr><tr><td>10</td><td>Rain</td><td>Mild</td><td>Normal</td><td>Weak</td><td>46</td></tr><tr><td>11</td><td>Sunny</td><td>Mild</td><td>Normal</td><td>Strong</td><td>48</td></tr><tr><td>12</td><td>Overcast</td><td>Mild</td><td>High</td><td>Strong</td><td>52</td></tr><tr><td>13</td><td>Overcast</td><td>Hot</td><td>High</td><td>Weak</td><td>44</td></tr><tr><td>14</td><td>Rain</td><td>Mild</td><td>High</td><td>Strong</td><td>30</td></tr></table>	Day`	Outlook	Temp	Humidity	Wind	Golf Players	1	Sunny	Hot	High	Weak	25	2	Sunny	Hot	High	Strong	30	3	Overcast	Hot	High	Weak	46	4	Rain	Mild	High	Weak	45	5	Rain	Cool	Normal	Weak	52	6	Rain	Cool	Normal	Strong	23	7	Overcast	Cool	Normal	Strong	43	8	Sunny	Mild	High	Weak	35	9	Sunny	Cool	Normal	Weak	38	10	Rain	Mild	Normal	Weak	46	11	Sunny	Mild	Normal	Strong	48	12	Overcast	Mild	High	Strong	52	13	Overcast	Hot	High	Weak	44	14	Rain	Mild	High	Strong	30	[20]
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Q4

CO-3; SO-2;
BL-3

- Predict the class label for a test sample ($A = 1, B = 1, C = 1$) using the naive Bayes approach
- Predict the class label for a test sample ($A = 1, B = 1, C = 1$) using the naive Bayes approach but with Laplace estimate.
- Compare the output from (a) and (b), which method is better and why?

[20]

Instance	A	B	C	Class
1	0	0	1	-
2	1	0	1	+
3	0	1	0	-
4	1	0	0	-
5	1	0	1	+
6	0	0	1	+
7	1	1	0	-
8	0	0	0	-
9	0	1	0	+
10	1	1	1	+

- For the one-dimensional data shown below, Classify the data point $x = 5.0$ according to its 1-, 3-, 5-, and 9-nearest neighbors (using majority vote)

X	0.5	3.0	4.5	4.6	4.9	5.2	5.3	5.5	7.0	9.5
Y	-	-	+	+	+	-	-	+	-	-

Q5

CO-2; SO-3;
BL-3

- Explain with Schematics how Logistic regression can be used for multi-class classification.
- What is hard margin and soft margin in SVM (use schematics)?
- Consider the training examples shown in the below Table for a binary classification problem.

[20]

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	-
4	F	F	4.0	+
5	F	T	7.0	-
6	F	T	3.0	-
7	F	F	8.0	-
8	T	F	7.0	+
9	F	T	5.0	-

- What is the entropy of this collection of training examples?
- What are the information gains of a_1 and a_2 relative to these training examples?
- For a_3 , which is a continuous attribute, compute the information gain for every possible split
- What is the best split (among a_1, a_2 , and a_3) according to the information gain?

Q6

CO-3; SO-2;
BL-3

a. Using Apriori algorithm for the below data, Find

- The frequent itemsets.
- Association rules for frequent itemsets with the largest number of items.

Assume that minimum support ($s = 2$) and minimum confident threshold ($c = 50\%$)

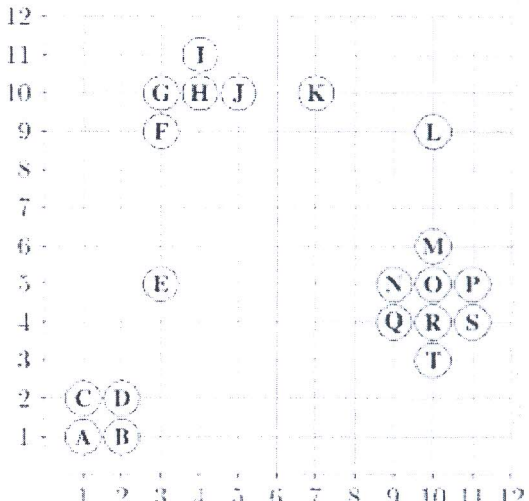
TID	ITEMSETS
T1	A,B
T2	B,D
T3	B,C
T4	A,B,D
T5	A,C
T6	B,C
T7	A,C
T8	A,B,C,E
T9	A,B,C

b. You are asked to evaluate the performance of two classification models, M1 and M2. The test set you have chosen contains 26 binary attributes, labeled as A through Z. The below table shows the posterior probabilities obtained by applying the models to the test set. (Only the posterior probabilities for the positive class are shown). As this is a two-class problem, $P(-) = 1 - P(+)$ and $P(-|A, \dots, Z) = 1 - P(+|A, \dots, Z)$.

Assume that we are mostly interested in detecting instances from the positive class:

- Plot the ROC curve for both M1 and M2. (You should plot them on the same graph.) Which model do you think is better? Explain your reasons.
- For model M1, suppose you choose the cutoff threshold to be $t = 0.1$. In other words, any test instances whose posterior probability is greater than t will be classified as a positive example. Compute the precision, recall, and F-measure for the model at this threshold value.
- Repeat the analysis using the same cutoff threshold on model M2. Compare the F-measure results for both models. Which model is better? Are the results consistent with what you expect from the ROC curve?

Instance	True Class	$P(+ A, \dots, Z, M_1)$	$P(+ A, \dots, Z, M_2)$
1	+	0.73	0.61
2	+	0.69	0.03
3	-	0.44	0.68
4	-	0.55	0.31
5	+	0.67	0.45
6	+	0.47	0.09
7	-	0.08	0.38
8	-	0.15	0.05
9	+	0.45	0.01

		10	-	0.35	0.04													
Q7 CO-3; SO-2; BL-3		a. Explain with Schematics two methods to overcome the initial randomization of the centroids in k-means clustering. b. For the below table, please perform single linkage agglomerative clustering and plot the dendrogram, showing the clusters for cluster size=3. <table><tr><th>Student_ID</th><th>Marks</th></tr><tr><td>1</td><td>10</td></tr><tr><td>2</td><td>7</td></tr><tr><td>3</td><td>28</td></tr><tr><td>4</td><td>20</td></tr><tr><td>5</td><td>35</td></tr></table> c. Measure the cluster cohesion (SSE) and cluster separation (SSB) for number of clusters=3, also get the SSE and SSB if we only have one cluster. d. For the below 2 dimensional feature space, please use DBSCAN to identify the core/border/noise points (use Manhattan distance as the distance function). Let MinPts = 2 and radius=1.1, show the clusters. 				Student_ID	Marks	1	10	2	7	3	28	4	20	5	35	[20]
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